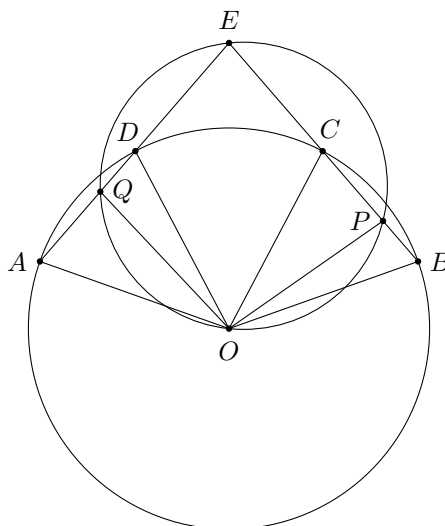




HK IMO Training 2023-2024 Problem Set 6 Suggested Solutions



Problem 21 (CentroAmerican MO 2014 Problem 2). Let $ABCD$ be a trapezoid inscribed in a circle with centre O , where AB and DC are parallel. The lines AD and BC meet at E . A circle passing through E and O meets the segments BC and AD at P and Q respectively. Prove that $BP = DQ$.



Solution. Note that $ABCD$ is an isosceles trapezoid. By symmetry, we have

$$\angle ODQ = \angle ODA = \angle OCB = \angle OBC = \angle OBP.$$

Together with $\angle OQD = \angle OPB$ and $OD = OB$, we deduce $\triangle ODQ \cong \triangle OBP$. Therefore, $DQ = BP$. □

Problem 22 (Turkey TST 2019 Problem 4). For any positive integer n with k digits, we say that a positive integer m is a *weak divisor* of n if m divides the number obtained by removing the leftmost a digits and the rightmost b digits of n , where a and b are nonnegative integers with $a + b < k$. For example, 19 is a weak divisor of 13579 because 19 divides 57, while 101 is a weak divisor of 2023 because 101 divides 202. Find all positive integers d such that there are finitely many positive integers n for which d is not a weak divisor of n .

Solution. Answer: All positive integers which are not divisible by 2 or 5.

If d is divisible by 2 or 5, then it is not a weak divisor of any positive integer of the form $11 \cdots 1$, because the unit digit is always 1 by removing some digits from it, and hence the resultant number is not divisible by 2 or 5.

If d is not divisible by 2 or 5, we claim that it is a weak divisor of all positive integers with at least $d + 1$ digits. Indeed, consider any integer $n = \overline{a_k \cdots a_2 a_1}$ with $k \geq d + 1$. Among the integers $\overline{a_1}, \overline{a_2 a_1}, \dots, \overline{a_{d+1} \cdots a_2 a_1}$, two of them must leave the same remainder when divided by d by the pigeonhole principle. Let $\overline{a_i \cdots a_2 a_1} \equiv \overline{a_j \cdots a_2 a_1} \pmod{d}$ where $i > j$. Then

$$\overline{a_i a_{i-1} \cdots a_{j+1} 00 \cdots 0} = \overline{a_i \cdots a_2 a_1} - \overline{a_j \cdots a_2 a_1} \equiv 0 \pmod{d}.$$

As $(d, 10) = 1$, this implies $d \mid \overline{a_i a_{i-1} \cdots a_{j+1}}$, and hence d is a weak divisor of n . □

Problem 23 (Turkey TST 2019 Problem 9). Let a, b, c be real numbers satisfying $b > 2c > 4a$ and

$$2(a^3 + b^3 + c^3) + 15(ab^2 + bc^2 + ca^2) > 16(a^2b + b^2c + c^2a) + 2abc.$$

Prove that $4a + b > 4c$.

Solution. Define $x = a - 2b$, $y = b - 2c > 0$ and $z = c - 2a > 0$. On the one hand, we have

$$\begin{aligned} xy^2 + yz^2 + zx^2 - 2xyz &= \sum_{\text{cyc}} (a - 2b)(b - 2c)^2 - 2(a - 2b)(b - 2c)(c - 2a) \\ &= -2 \sum_{\text{cyc}} a^3 + 16 \sum_{\text{cyc}} a^2b - 15 \sum_{\text{cyc}} ab^2 + 2abc \\ &< 0. \end{aligned}$$

On the other hand, we have

$$xy^2 + yz^2 + zx^2 - 2xyz = z \left(x - y + \frac{y^2}{2z} \right)^2 + y(y - z)^2 + \frac{y^2(4z^2 - y^2)}{4z}.$$

Since this is less than 0 and $y, z > 0$, we must have $4z^2 < y^2$, i.e. $2z < y$. It follows that $2(c - 2a) < b - 2c$, or $4a + b > 4c$. $\square$

Problem 24 (Serbian MO 2017 Problem 3). Let n be a positive integer. There are $2n - 1$ coins lying in a row. Initially, only the middle coin (the n th coin) has its head facing upwards, while all other coins have their tails facing upwards. In each step, we can choose at least three consecutive coins such that the leftmost and the rightmost coins have their tails facing upwards and all other coins have their heads facing upwards, and we flip all these coins. At most how many steps can be performed?

Solution. Answer: $\left\lfloor \frac{2^{n+1} - 5}{3} \right\rfloor$.

We use a binary string to represent the coins, where 1 represents a coin with its tail facing upwards, and 0 represents a coin with its head facing upwards. Initially, the binary string $a_1a_2 \cdots a_n \cdots a_{2n-2}a_{2n-1}$ is $11 \cdots 0 \cdots 11$. Each time we can flip a string from $100 \cdots 01$ to $011 \cdots 10$.

Define

$$S = a_n + (2a_{n-1} + 2^2a_{n-2} + \cdots + 2^{n-1}a_1) + (2a_{n+1} + 2^2a_{n+2} + \cdots + 2^{n-1}a_{2n-1}).$$

Consider a single step.

- If we flip a string $a_i \cdots a_j$ where $i \geq n$, then S is decreased by

$$2^{i-n} + 2^{j-n} - (2^{i+1-n} + 2^{i+2-n} + \cdots + 2^{j-1-n}) = 3 \cdot 2^{i-n} \geq 3.$$

- Similarly, if we flip a string $a_i \cdots a_j$ where $j \leq n$, then S is decreased by $3 \cdot 2^{n-j} \geq 3$.
- If we flip a string $a_i \cdots a_j$ where $i < n < j$, then S is decreased by

$$2^{n-i} - (2^{n-i-1} + \cdots + 2^2 + 2) - 1 - (2 + 2^2 + \cdots + 2^{j-1-n}) + 2^{j-n} = 3.$$

This shows S is decreased by at least 3 in each step. Since S is always positive (as a new 1 is generated in each step), and the initial value of S is $2(2 + 2^2 + \cdots + 2^{n-1}) = 2^{n+1} - 4$, at most $\left\lfloor \frac{2^{n+1} - 5}{3} \right\rfloor$ steps can be carried out.

It remains to show that there is a way to flip the coins such that S is decreased by 3 in each step. Initially, we have a string of the form $A101\bar{A}$, where $\bar{A}$ is the string obtained by writing the digits of A in the reverse order.

- We flip the middle string 101. The string becomes $A010\bar{A}$, and S is decreased by 3.
- If A is the zero string, we are done because $S = 1$ (which cannot be further reduced). Otherwise, A contains a digit 1. We flip the string where $a_n = 1$ is the rightmost digit. The string becomes $B100\bar{A}$ where B is obtained from A by changing the last digits $100 \cdots 0$ (possibly without 0) to $011 \cdots 1$. For simplicity, we write $B = f(A)$. From above, S is decreased by 3.
- Since $\bar{A}$ contains a digit 1, we can flip the string where $a_{n-1} = 1$ is the leftmost digit. The string becomes $B011\bar{B}$, and S is decreased by 3.
- If B is the zero string, we are done because $S = 3$. Otherwise, we flip the string where $a_n = 1$ is the rightmost digit. The string becomes $C101\bar{B}$ where $C = f(B)$, and S is decreased by 3.
- We flip the middle string 101. The string becomes $C010\bar{B}$, and S is decreased by 3.
- We flip the string where $a_n = 1$ is the leftmost digit. The string becomes $C001\bar{C}$, and S is decreased by 3.
- If C is the zero string, we are done because $S = 2$. Otherwise, we flip the string where $a_{n+1} = 1$ is the rightmost digit. The string becomes $D110\bar{C}$ where $D = f(C)$, and S is decreased by 3.
- We flip the string where $a_n = 1$ is the leftmost digit. The string becomes $D101\bar{D}$, and S is decreased by 3.

Note that we go back to a similar state as the initial one. Therefore, by repeating the procedure, we must eventually end. As justified, S is decreased by 3 in every step. $\square$